

World Geodetic Survey 1984

MEAN EQUATORIAL RADIUS

**	r_e, a_e	1.0 DU	=	6378.137000000 km (± 2 m)
			=	20925646.325459318 ft
			=	3963.190591943 miles
			=	3443.918466523 NM

TIME

		1.0 TU	=	13.44685115881 solar min
			=	806.81109528434 solar sec
			=	0.0093380913806 solar days
			=	0.0093636582283 sidereal days
θ_{go}	1 Jan 2021 0 UT		=	100.86875630°
θ_{go}	1 Jan 2022 0 UT		=	100.63004655°
θ_{go}	1 Jan 2023 0 UT		=	100.39133688°
θ_{go}	1 Jan 2024 0 UT		=	100.15262729°
θ_{go}	1 Jan 2025 0 UT		=	100.89956515°

SPEED

	1.0 DU/TU	=	7.90536629614902 km/solar sec
		=	25936.240153073901 ft/solar sec

GRAVITATIONAL PARAMETER

**	μ	1.0 DU ³ /TU ²	=	398600.5 km ³ /(solar sec) ² (± 0.03) includes mass of atmosphere
			=	14076443812518712.8 ft ³ /(solar sec) ²

EARTH ROTATION

**	$w_{\hat{A}}$		=	0.00007292115 rad/solar sec ($\pm 1.5 \times 10^{-12}$) wrt Precess Frame
			=	0.25068447733746215 deg/solar min
			=	6.30038736 rad/solar day
			=	0.058833592900857 rad/TU

SHAPE

+		Mass	=	5.9733328 x 10 ²⁴ kg
+	b_e	Semi-Minor Axis	=	6356.7523142 km
+	e_e	Eccentricity of Earth	=	0.0818191908426
+	$(e_e)^2$		=	0.00669437999013
+	f	Flattening of Earth	=	1.0 / 298.257223563
**	J_2		=	0.00108263
	J_3		=	-0.00000254
	J_4		=	-0.00000161

CONVERSIONS & CONSTANTS

1.0 Ft	=	0.0003048 km
1.0 NM	=	1.852 km (exact)
1.0 Mile	=	1.609344 km
1.0 Mile	=	5280 ft (exact)
1.0 solar day	=	1.002737791737697 sidereal days
1.0 radian	=	57.2957795130823°
π	=	3.14159265358979

SUN CONSTANTS

1.0 DU $_{\odot}$	=	696265 km
1.0 AU	=	149597870 km
1.0 TU $_{\odot}$	=	83710.714905 solar min
	=	58.1324409 solar day
1.0 AU/TU	=	29.7846916749 km/sec
$\mu_{\odot}$	=	1.32712438 x 10 ¹¹ km ³ /sec ²
Mass $_{\odot}$	=	1.9891 x 10 ³⁰ kg

** indicates official parameter from WGS-84
+ indicates derived parameter from WGS-84

DAY NUMBER FOR CALENDAR YEAR

PERPETUAL

Day	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Day
1	001	032	060	091	121	152	182	213	244	274	305	335	1
2	002	033	061	092	122	153	183	214	245	275	306	336	2
3	003	034	062	093	123	154	184	215	246	276	307	337	3
4	004	035	063	094	124	155	185	216	247	277	308	338	4
5	005	036	064	095	125	156	186	217	248	278	309	339	5
6	006	037	065	096	126	157	187	218	249	279	310	340	6
7	007	038	066	097	127	158	188	219	250	280	311	341	7
8	008	039	067	098	128	159	189	220	251	281	312	342	8
9	009	040	068	099	129	160	190	221	252	282	313	343	9
10	010	041	069	100	130	161	191	222	253	283	314	344	10
11	011	042	070	101	131	162	192	223	254	284	315	345	11
12	012	043	071	102	132	163	193	224	255	285	316	346	12
13	013	044	072	103	133	164	194	225	256	286	317	347	13
14	014	045	073	104	134	165	195	226	257	287	318	348	14
15	015	046	074	105	135	166	196	227	258	288	319	349	15
16	016	047	075	106	136	167	197	228	259	289	320	350	16
17	017	048	076	107	137	168	198	229	260	290	321	351	17
18	018	049	077	108	138	169	199	230	261	291	322	352	18
19	019	050	078	109	139	170	200	231	262	292	323	353	19
20	020	051	079	110	140	171	201	232	263	293	324	354	20
21	021	052	080	111	141	172	202	233	264	294	325	355	21
22	022	053	081	112	142	173	203	234	265	295	326	356	22
23	023	054	082	113	143	174	204	235	266	296	327	357	23
24	024	055	083	114	144	175	205	236	267	297	328	358	24
25	025	056	084	115	145	176	206	237	268	298	329	359	25
26	026	057	085	116	146	177	207	238	269	299	330	360	26
27	027	058	086	117	147	178	208	239	270	300	331	361	27
28	028	059	087	118	148	179	209	240	271	301	332	362	28
29	029		088	119	149	180	210	241	272	302	333	363	29
30	030		089	120	150	181	211	242	273	303	334	364	30
31	031		090		151		212	243		304		365	31

FOR LEAP YEARS ONLY (USE IN 2016, 2020, 2024, etc)

Day	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Day
1	001	032	061	092	122	153	183	214	245	275	306	336	1
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10	010	041	070	101	131	162	192	223	254	284	315	345	10
11	011	042	071	102	132	163	193	224	255	285	316	346	11
12	012	043	072	103	133	164	194	225	256	286	317	347	12
13	013	044	073	104	134	165	195	226	257	287	318	348	13
14	014	045	074	105	135	166	196	227	258	288	319	349	14
15	015	046	075	106	136	167	197	228	259	289	320	350	15
16	016	047	076	107	137	168	198	229	260	290	321	351	16
17	017	048	077	108	138	169	199	230	261	291	322	352	17
18	018	049	078	109	139	170	200	231	262	292	323	353	18
19	019	050	079	110	140	171	201	232	263	293	324	354	19
20	020	051	080	111	141	172	202	233	264	294	325	355	20
21	021	052	081	112	142	173	203	234	265	295	326	356	21
22	022	053	082	113	143	174	204	235	266	296	327	357	22
23	023	054	083	114	144	175	205	236	267	297	328	358	23
24	024	055	084	115	145	176	206	237	268	298	329	359	24
25	025	056	085	116	146	177	207	238	269	299	330	360	25
26	026	057	086	117	147	178	208	239	270	300	331	361	26
27	027	058	087	118	148	179	209	240	271	301	332	362	27
28	028	059	088	119	149	180	210	241	272	302	333	363	28
29	029	060	089	120	150	181	211	242	273	303	334	364	29
30	030		090	121	151	182	212	243	274	304	335	365	30
31	031		091		152		213	244		305		366	31

ASTRO ENGR 321 CONSTANTS AND EQUATIONS

Constants

$$\omega_{\oplus} = 15 \text{ deg/sidereal hr}$$

$$\rho_0 = 1.225 \text{ kg/m}^3$$

$$\beta = \frac{1}{7.315 \text{ km}}$$

Satellites - The Basics

$$\ddot{\mathbf{R}} + \frac{\mu}{R^3} \mathbf{R} = \mathbf{0} \quad R = \frac{a(1-e^2)}{1+e \cos(v)}$$

$$h = RV \cos(\phi) = \sqrt{\mu a(1-e^2)} \quad \bar{h} = \bar{R} \times \bar{V}$$

$$a = \frac{R_a + R_p}{2} \quad \mathbb{P} = 2\pi \sqrt{\frac{a^3}{\mu}}$$

$$R_p = a(1-e) \quad R_a = a(1+e)$$

$$\bar{e} = \frac{\bar{V} \times \bar{h}}{\mu} - \frac{\bar{R}}{R} \quad p = a(1-e^2) = \frac{h^2}{\mu}$$

$$e = \frac{R_1 - R_2}{R_2 \cos(v_2) - R_1} \quad \varepsilon = \frac{V^2}{2} - \frac{\mu}{R} = \frac{-\mu}{2a}$$

$$\tan(\phi) = \frac{e \sin(v)}{1+e \cos(v)} \quad a^3 n^2 = \mu$$

Time

$$\theta_g = \theta_{go} + \frac{\text{solar day}}{\text{sidereal day}} * 2\pi \text{ (or } 360^\circ) * D$$

$$D = \text{Day\#} - 1 + \left(\frac{HR}{24}\right) + \left(\frac{MIN}{1440}\right) + \left(\frac{SEC}{86400}\right)$$

where D is Elapsed days

$$LST = \theta_g + \lambda_E$$

$$MST = UT - 7 \text{ hrs} \quad MDT = UT - 6 \text{ hrs}$$

Preliminary Orbit Determination

$$x = \left| \frac{a_e}{\sqrt{1-e^2 \sin^2(L)}} + H \right| \cos(L)$$

$$z = \left| \frac{a_e(1-e^2)}{\sqrt{1-e^2 \sin^2(L)}} + H \right| \sin(L)$$

$$\bar{R}_{site} = x \cos(LST) \hat{i} + x \sin(LST) \hat{j} + z \hat{k}$$

$$\rho_S = -\rho \cos(Az) \cos(El)$$

$$\rho_E = \rho \sin(Az) \cos(El)$$

$$\rho_Z = \rho \sin(El)$$

$$\dot{\rho}_S = -\dot{\rho} \cos(Az) \cos(El) + \rho \dot{A}z \sin(Az) \cos(El) + \rho \dot{E}l \cos(Az) \sin(El)$$

$$\dot{\rho}_E = \dot{\rho} \sin(Az) \cos(El) + \rho \dot{A}z \cos(Az) \cos(El) - \rho \dot{E}l \sin(Az) \sin(El)$$

$$\dot{\rho}_Z = \dot{\rho} \sin(El) + \rho \dot{E}l \cos(El)$$

$$\begin{bmatrix} X_I \\ X_J \\ X_K \end{bmatrix} = [Rot_3(-LST)][Rot_2(-COLAT)] \begin{bmatrix} X_S \\ X_E \\ X_Z \end{bmatrix}$$

$$\bar{R} = \bar{\rho} + \bar{R}_{site} \quad \bar{V} = \dot{\bar{\rho}} + \bar{\omega}_{\oplus} \times \bar{R}$$

$$\begin{aligned} \mathbf{V} = & -\Delta t_{32} \left(\frac{1}{\Delta t_{21} \Delta t_{31}} + \frac{\mu}{12 R_3^3} \right) \bar{R}_1 + \\ & (\Delta t_{32} - \Delta t_{21}) \left(\frac{1}{\Delta t_{21} \Delta t_{32}} + \frac{\mu}{12 R_2^3} \right) \bar{R}_2 + \\ & \Delta t_{21} \left(\frac{1}{\Delta t_{32} \Delta t_{31}} + \frac{\mu}{12 R_3^3} \right) \bar{R}_3 \end{aligned}$$

Transfers

$$\Delta V_{\text{continuous}} = |V_1 - V_2|$$

$$\Delta V^2 = V_1^2 + V_2^2 - 2V_1 V_2 \cos(\Delta\theta)$$

$$\Delta\theta = \Delta i \text{ (Combined Plane Change),}$$

$$\Delta\phi \text{ (Non Tangent Burn),}$$

$$\delta = 180^\circ - \cos^{-1} \left(\frac{V_1^2 + \Delta V^2 - V_2^2}{2V_1 \Delta V} \right)$$

Rendezvous

$$TOF = \pi \sqrt{\frac{a^3}{\mu}} \quad \omega = \sqrt{\frac{\mu}{R_{\text{circ}}^3}}$$

$$\alpha_{\text{Lead}} = \omega_{\text{Target}} \times TOF \quad \phi_{\text{Final}} = \pi - \alpha_{\text{Lead}}$$

$$\text{Wait Time}_{\text{Coplanar}} = \frac{\phi_f - \phi_i (\pm 2\pi n)}{\omega_{\text{Target}} - \omega_{\text{Interceptor}}}$$

$$\text{Wait Time}_{\text{NonCoplanar}} = \frac{\phi_f - \phi_i + 2\pi n}{\omega_{\text{Target}}} = \frac{\alpha_i - \alpha_f + 2\pi n}{\omega_{\text{Target}}}$$

$$TOF_{\text{tgt}} = TOF_{\text{int to node}} + \mathbb{P}_{\text{phase}} + TOF_{\text{Hohmann}}$$

Kepler's Problem

$$M = E - e \sin(E)$$

$$\cos(E) = \frac{e + \cos(v)}{1 + e \cos(v)} \quad \cos(v) = \frac{\cos(E) - e}{1 - e \cos(E)}$$

$$\tan\left(\frac{v}{2}\right) = \sqrt{\frac{1+e}{1-e}} \tan\left(\frac{E}{2}\right)$$

$$E_{n+1} = E_n + \frac{M - E_n + e \sin(E_n)}{1 - e \cos(E_n)}$$

COEs to R and V

$$\bar{R} = R[\cos(v) \hat{P} + \sin(v) \hat{Q}]$$

$$\bar{V} = \sqrt{\frac{\mu}{p}} [-\sin(v) \hat{P} + (e + \cos(v)) \hat{Q}]$$

$$\begin{bmatrix} X_I \\ X_J \\ X_K \end{bmatrix} = [Rot_3(-\Omega)][Rot_1(-i)][Rot_3(-\omega)] \begin{bmatrix} X_P \\ X_Q \\ X_W \end{bmatrix}$$

Perturbations

$$\bar{n} = n_0 \left[1 + \frac{3}{2} J_2 \left(\frac{R_{\oplus}}{p_0} \right)^2 \sqrt{1 - e_0^2} \left(1 - \frac{3}{2} \sin^2(i_0) \right) \right]$$

$$n = n_0 + \dot{n}_0 \Delta t$$

$$\dot{e}_{\text{drag}} = \frac{-2(1 - e_0) \dot{n}_0}{3n_0}$$

$$e = e_0 + \dot{e}_{\text{drag}} \Delta t$$

$$\dot{\Omega}_{J_2} = \left[-\frac{3}{2} J_2 \left(\frac{R_{\oplus}}{p_0} \right)^2 \cos(i_0) \right] \bar{n}$$

$$\Omega = \Omega_0 + \dot{\Omega}_{J_2} \Delta t$$

$$\dot{\omega}_{J_2} = \left[\frac{3}{2} J_2 \left(\frac{R_{\oplus}}{p_0} \right)^2 \left(2 - \frac{5}{2} \sin^2(i_0) \right) \right] \bar{n}$$

$$\omega = \omega_0 + \dot{\omega}_{J_2} \Delta t$$

$$M = M_0 + n_0 \Delta t + \frac{\dot{n}_0}{2} \Delta t^2$$

$$\bar{a}_{\text{drag}} = -\frac{1}{2} r_0 \frac{C_D A}{m} V \bar{V}$$

ASTRO ENGR 321 CONSTANTS AND EQUATIONS

Entry

$$\dot{R} = V \sin(\phi_E)$$

$$\dot{V} = -\frac{\rho_o}{2\Delta} V^2 e^{-\beta h}$$

$$\Delta = \frac{m}{C_D A}$$

$$V = V_E e^{\left[\frac{\rho_o}{2\Delta\beta \sin(\phi_E)} e^{-\beta h} \right]}$$

$$V_{Max \frac{\dot{V}}{g}} = V_E e^{-.5} \approx 0.61 V_E$$

$$h_{Max \frac{\dot{V}}{g}} = \frac{1}{\beta} \ln \left(-\frac{\rho_o}{\Delta\beta \sin(\phi_E)} \right)$$

$$-\frac{\dot{V}}{g_{Max}} = \frac{\beta V_E^2 \sin(\phi_E)}{2g_0 e^1}$$

Proximity Operations

$$x(t) = \frac{\dot{x}_0}{n} \sin(nt) - \left(3x_0 + \frac{2\dot{y}_0}{n} \right) \cos(nt) + \left(4x_0 + \frac{2\dot{y}_0}{n} \right)$$

$$\dot{x}(t) = \dot{x}_0 \cos(nt) + (3nx_0 + 2\dot{y}_0) \sin(nt)$$

$$y(t) = \left(6x_0 + \frac{4\dot{y}_0}{n} \right) \sin(nt) + \frac{2\dot{x}_0}{n} \cos(nt) - (6nx_0 + 3\dot{y}_0)t + \left(y_0 - \frac{2\dot{x}_0}{n} \right)$$

$$\dot{y}(t) = (6nx_0 + 4\dot{y}_0) \cos(nt) - 2\dot{x}_0 \sin(nt) - (6nx_0 + 3\dot{y}_0)$$

$$z(t) = z_0 \cos(nt) + \frac{\dot{z}_0}{n} \sin(nt)$$

$$\dot{z}(t) = -z_0 n \sin(nt) + \dot{z}_0 \cos(nt)$$

Matrix Form:

$$\begin{bmatrix} \bar{r}_{rel} \\ \bar{v}_{rel} \end{bmatrix} = \Phi \begin{bmatrix} \bar{r}_{rel0} \\ \bar{v}_{rel0} \end{bmatrix}$$

Where:

$$\Phi = \begin{bmatrix} 4 - 3C & 0 & 0 & S/n & 2(1 - C)/n & 0 \\ 6S - 6nt & 1 & 0 & 2(C - 1)/n & -3t + 4S/n & 0 \\ 0 & 0 & C & 0 & 0 & S/n \\ 3nS & 0 & 0 & C & 2S & 0 \\ 6n(C - 1) & 0 & 0 & -2S & 4C - 3 & 0 \\ 0 & 0 & -nS & 0 & 0 & C \end{bmatrix}$$

$$C = \cos(nt)$$

$$S = \sin(nt)$$

Targeting

$$\begin{bmatrix} \bar{r}_{rel} \\ \bar{v}_{rel} \end{bmatrix} = \begin{bmatrix} \Phi_{rr} & \Phi_{vr} \\ \Phi_{rv} & \Phi_{vv} \end{bmatrix} \begin{bmatrix} \bar{r}_{rel0} \\ \bar{v}_{rel0} \end{bmatrix}$$

$$\bar{v}_{rel0} = \Phi_{vr}^{-1} [\bar{r}_{rel} - \Phi_{rr} \bar{r}_{rel0}]$$

Misc.

$$ROT_1(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & \sin(\alpha) \\ 0 & -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

$$ROT_2(\beta) = \begin{bmatrix} \cos(\beta) & 0 & -\sin(\beta) \\ 0 & 1 & 0 \\ \sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$

$$ROT_3(\gamma) = \begin{bmatrix} \cos(\gamma) & \sin(\gamma) & 0 \\ -\sin(\gamma) & \cos(\gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$